

DATA SCIENTIST

Shivani Avasarala

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Data Scientist with over 5 years of experience leveraging advanced machine learning algorithms, predictive modeling, and statistical techniques to drive impactful business solutions. Skilled in Text Analytics, Statistical Machine Learning, and Data Mining, with a strong track record of developing data-driven recommendations across diverse industries. Proficient in Python, SQL, and Excel, with hands-on experience in visualizing insights through Power BI and Tableau. Extensive experience transforming complex data into actionable business insights using statistical analysis and visualization techniques. Comprehensive understanding of the Data Science project lifecycle, including data collection, storage, and visualization for large, complex datasets. Experienced in creating ER diagrams and schemas to support efficient data management and engineering pipelines.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Algorithms: Linear Regression, Logistic regression, Naïve Bayes, Decision Trees, Random Forest, Boosting Algorithms, Deep Neural Networks, CNNs, RNNs, Transformers, BERT

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Pytorch, TensorFlow, Hugging Face

Toolkits: Jupyter, Power BI, Tableau

WORK EXPERIENCE

Motorola Solutions, Chicago IL

Jan 2024- Till Date

Data Scientist

Classification of Customer Reviews:

- Developed a multi-class classification system using ensemble methods on **Azure Machine Learning**, achieving 88% accuracy.
- Built **sentiment analysis models** using **BERT** on product reviews, automating sentiment classification with 88% accuracy.
- Uncovered 3 critical product issues through K-means clustering of customer feedback, enabling proactive resolution.
- Implemented a **Retrieval-Augmented Generation (RAG)** system using **GPT-4**, enabling context-aware customer review processing with 92% relevancy score.
- Designed and deployed **Azure Data Factory-based ETL pipelines** to preprocess customer reviews, ensuring **scalable data ingestion, cleansing, and transformation** before analysis.

Market Basket Analysis:

- Spearheaded development of next-purchase prediction engine with 78% accuracy for top 5 recommendations
- Reduced feature dimensionality from 100+ attributes while retaining 95% of variance using PCA and t-SNE.
- Developed **Decision Tree** models to predict the next likely purchase, achieving 78% accuracy for top 5 recommendations.
- Created a feature engineering pipeline using pandas to process transaction data, including time-based aggregations and customer segmentation.
- Created **automated outlier detection** to identify and handle pricing anomalies, improving model robustness.
- Deployed models using **Azure Kubernetes Service (AKS)** for dynamic scaling and efficient resource utilization.

Supply Chain Optimization

- Developed ensemble learning models for 12-month value predictions.
- Implemented a multi-level forecasting system incorporating SKU-level predictions with time series decomposition, store-level aggregations with hierarchical models, and seasonal adjustments using STL decomposition.
- Built **custom optimization algorithms** using **linear programming** for inventory allocation across 200+ locations.

Saber Health Group, Manhattan NY

Oct 2022 – Sep 2023

Data Scientist

Disease progression and Treatment recommendation system:

- Built Lasso and Ridge Regression Models for chronic disease progression prediction.
- **Engineered time-dependent features** to account for periodic health measurements, leading to a 15% increase in prediction accuracy.
- Implemented data cleansing and data analysis workflows using Google Dataflow based ETL pipelines, to handle missing medical records, normalizing complex medical terminologies, and resolving inconsistent patient identifiers
- Automated report generation using Python and **PowerBI** to create interactive dashboards and weekly insights on treatment effectiveness across patient cohorts, enabling clinical teams to make data-driven decisions.
- Performed statistical analysis and A/B testing to evaluate treatment effectiveness, presenting results through interactive PowerBI visualizations with custom **DAX** measures.

TATARIA LIVING, Mumbai, India.

March 2020 - June 2022

Data Analyst

- Created and maintained SQL queries for operational and supply chain data within **Google BigQuery**, ensuring **data accuracy** and consistency for analysis.
- Performed in-depth exploratory data analysis using SQL and Excel to identify patterns in inventory movements, leading to optimization of stock levels and 30% reduction in stockouts.
- Developed automated **data quality checks** using SQL to detect anomalies and inconsistencies in supply chain data, significantly enhancing data reliability for downstream reporting and analysis.
- Leveraged advanced SQL techniques, including complex joins, window functions, and subqueries, to extract actionable insights from interconnected datasets.
- Built **PowerBI dashboards** with drill-through capabilities for tracking key metrics.
- Developed VBA-enabled Excel models, automating routine reporting and reducing generation time by 40%.

Capstone:

NOSWARM/ Northeastern University, Boston, MA.

Sept 2023- Dec 2023

Business Analyst Advisor

- Led a project using 400 drone images, testing **YOLOv8** and **YOLO-NAS** for object detection on Google Colab.
- Analyzed monocular/stereo vision for distance estimation and used centroid tracking for velocity calculation.

EDUCATION

- **Masters in business analytics**, Northeastern University, Boston | GPA: 3.6
- **Bachelors in Business Administration**, ICAI Foundation of Higher Education, India | GPA: 3.2